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Autoregression Explained via Partial Correlation

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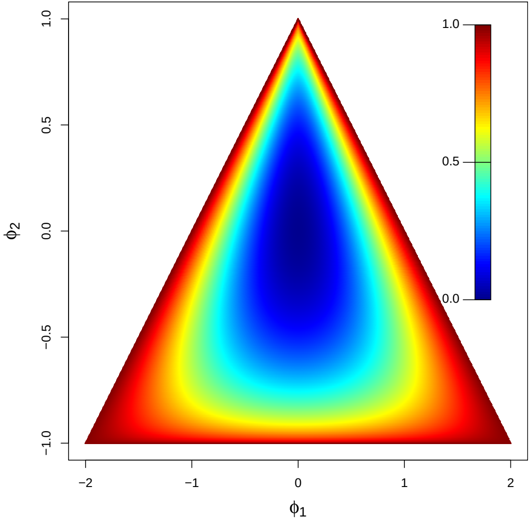
Time Series Group

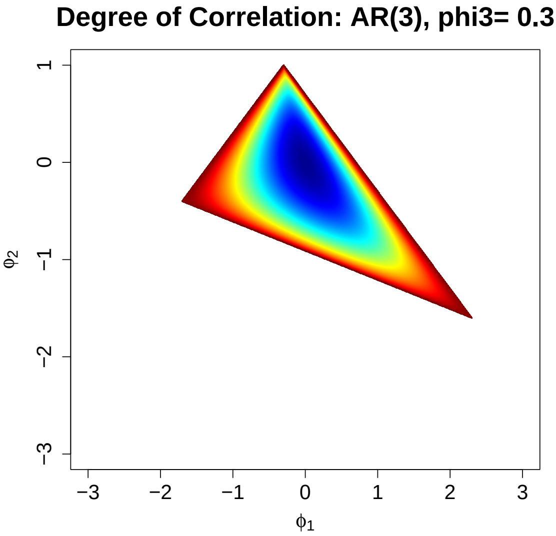
Aim

To give more intuition on causality and motivate better time series simulation studies.

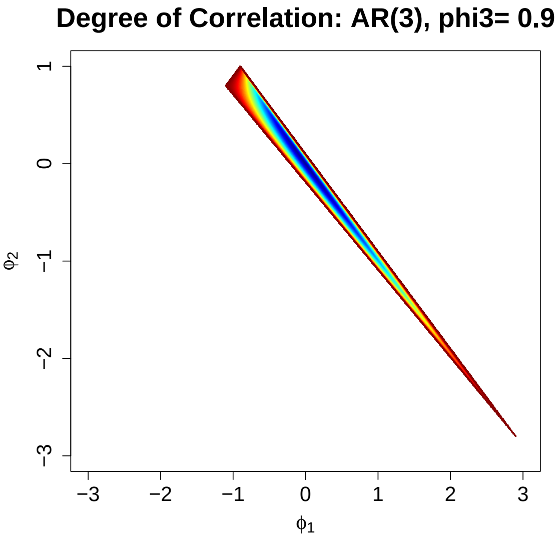
Solution AR2



Solution AR3



Solution AR3



Problem - Causality

AR(p):

$$Y_t = \sum_{j=1}^{p} \phi_j Y_{t-j} + \epsilon_t \quad t = 0, \pm 1, \pm 2, \dots,$$

Causal:

$$Y_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j} \quad \sum_{j=0}^{\infty} |c_j| < \infty;$$

How to check causality?

Problem - Causality

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How to check causality?

Roots of the polynomial:

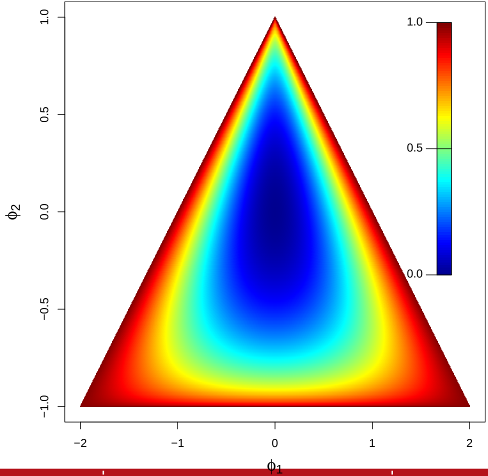
$$\Phi(x) = 1 - \phi_{p,1} x - \phi_{p,2} x^2 - \cdots \phi_{p,p} x^p$$

Are the roots outside the unit circle?

AR1: $\phi_{1,1} \in (-1, 1)$

AR2:

What about higher orders?



A different way

PACF at lag h: $\alpha(h)$.

$$|\alpha(h)| < 1 \quad \text{for} \quad 1 \leq h \leq p \quad \text{and} \quad \alpha(h) = 0 \quad \text{for} \quad h > p.$$

Ramsey (1974) noted a 1:1 mapping. Any point in $(-1, 1)^p$ corresponds to a unique point in the causal parameter space:

$$(\phi_{p,1}, \ldots, \phi_{p,p}) = g\left(\alpha(1), \ldots, \alpha(p)\right)$$

$$(\alpha(1), \ldots, \alpha(p)) = g^{-1}\left(\phi_{p,1}, \ldots, \phi_{p,p}\right).$$

Durbin-Levinson

The map g can be described via the Durbin-Levinson algorithm, which provides

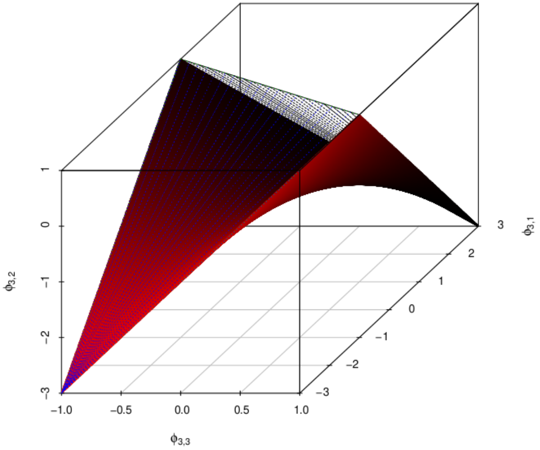
$$\begin{aligned} \phi_{k,k} &= \alpha(k) \\ \phi_{k,j} &= \phi_{k-1,j} - \phi_{k,k}\phi_{k-1,k-j} \qquad k > 1, 1 < j < k-1. \end{aligned}$$

Equivalent approach

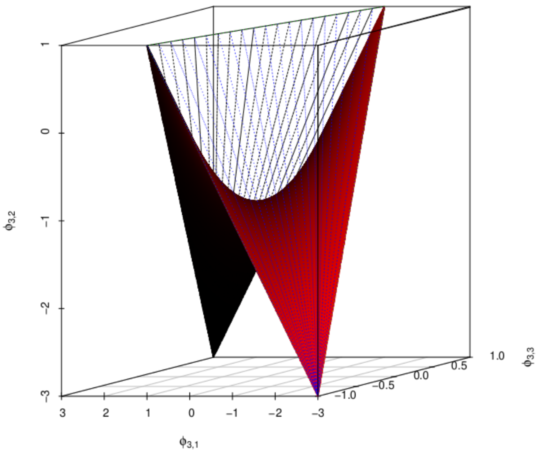
Let R_p denote the causal parameter space for an order p AR. Any of the following conditions can be used to assess if $\phi \in R_p$.

1. The roots of $\Phi(x) = 1 - \phi_{p,1}x - \phi_{p,2}x^2 - \cdots \phi_{p,p}x^p$ are outside the unit circle.
2. $|\phi_{h,h}| < 1$ for $1 \leq h \leq p$, and for each $k = 2, \ldots, p$ and $j = 1, \ldots, k$, $\phi_{k,j} = \phi_{k-1,j} - \phi_{k,k}\phi_{k-1,k-j}$.
3. $|\phi_{p,p}| < 1$, $\phi_{2i,2i} > -1$ for $i < p/2$, and for each $k = 2, \ldots, p$ and $j = 1, \ldots, k$,
$$\sum_{j=1}^{k} \phi_{k,j} < 1 \quad \text{and} \quad \sum_{j=1}^{k} (-1)^{j+1}\phi_{k,j} > -1,$$
with $\phi_{k,j} = \phi_{k-1,j} - \phi_{k,k}\phi_{k-1,k-j}$.
4. For $p > 1$, $(\phi_{p-1,1}, \ldots, \phi_{p-1,p-1}) \in R_{p-1}$, $|\phi_{p,p}| < 1$ and $\phi_{p,j} = \phi_{p-1,j} - \phi_{p,p}\phi_{p-1,p-j}$ for $j = 1, \ldots, p-1$.

Causal Region AR3



Causal Region AR3



Causal Region AR4

Dropbox Link

So what?

Need to summarise: $R_n(i, j) = corr(Y_i, Y_j)$, for $1 \leq i, j \leq n$

Measure of total correlation:

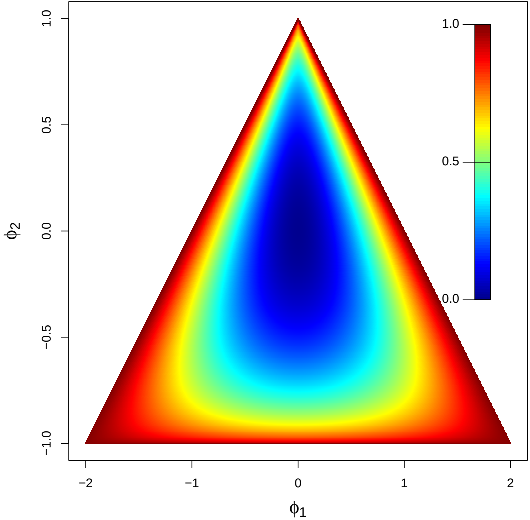
$$\rho(\{Y_t\}) = 1 - \det(R_n) = 1 - \prod_{k=1}^{p}(1 - \alpha(k)^2)^{p+1-k}.$$

where $\det$ denotes the determinant.

$\rho(\{Y_t\}) \to 1$, high correlation

$\rho(\{Y_t\}) \to 0$, low correlation

AR2 Total Correlation



Simulation Studies

1. Generate $\{X_i\}_{i\in(1,\dots,p)}$ independently having, $0 \leq \ell, u \leq 1$ Uniform density
$$f(x) = \frac{1}{(u-\ell)}\mathbb{I}_{\ell<x<u}.$$
2. Generate $\{R_i\}$ independent of $\{X_i\}$, $P(R=1) = P(R=-1) = 0.5$.
3. Let each $\alpha(i) = R_iX_i$ and solve for $\{\phi_{p,j}\}$.

Larger values of (ℓ, u) for one or more i correspond to more correlation.

AR(3)

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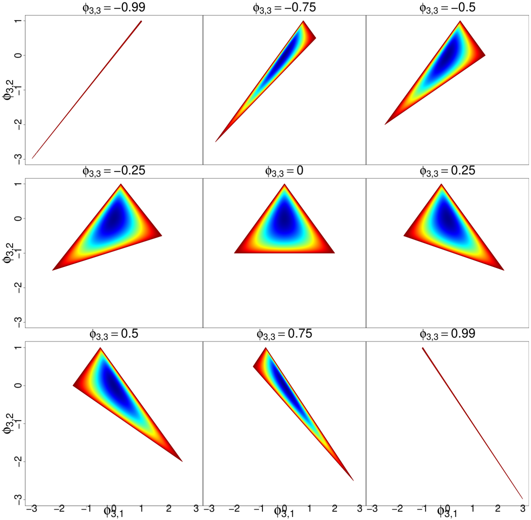
$\ell = 0.8$, $u = 0.9$, 10,000 simulations, $\rho(\{Y_t\}) > 0.9$, in 90% of simulations.



Summary

- Showed how PACF and AR coefficients are linked
- Used this link to describe causality regions in higher dimensions
- Visualized up to AR(4)
- Utilized the new description for simulation study generation

Code will be embedded in the `GRATIS: GeneRAting TIme Series with diverse and controllable characteristics` R package.

Email Rebecca for a script in the meantime.